

Sharp boundary bounds and edit instability in uniquely dominated bipartite graphs

5 October 2026

Third version

Abstract

We determine the sharp edge bounds for bipartite graphs without isolated vertices that have a unique minimum dominating set of size $\gamma \geq 2$ and order $3\gamma + 1$ or $3\gamma + 2$. The maxima are $\gamma(\gamma + 7)/2$ and $\lceil \gamma^2/2 \rceil + 5\gamma$, respectively, and all equality graphs are classified. There is one equality class at the first order. At the second order there are three classes when $\gamma = 2$, one for odd $\gamma \geq 3$, and two for even $\gamma \geq 4$. We then study edge-edit distance, minimized over all vertex relabellings, at order $3\gamma + 1$. A graph with edge deficit t is within $12\gamma t$ edits of the unique extremal graph. Connected examples require $\Omega(\gamma t)$ edits for $1 \leq t \leq \gamma/2 - 1$ when γ is even. In particular, relative edge deficit can tend to zero while normalized edit distance stays bounded away from zero. The proofs are elementary, using private-neighbour replacements, local edge deficits, and explicit near-extremal constructions. Previously reported finite counterexamples and construction subfamilies are recovered and explicitly credited.

Keywords: unique minimum dominating set; bipartite graph; extremal size; private neighbour.

2020 Mathematics Subject Classification: 05C69, 05C35.

1 Introduction

All graphs in this paper are finite and simple. A set $D \subseteq V(G)$ is *dominating* if every vertex outside D has a neighbour in D . The domination number $\gamma(G)$ is the minimum cardinality of a dominating set. A graph has a *unique minimum dominating set* if exactly one dominating set has cardinality $\gamma(G)$. Throughout, uniqueness refers to minimum cardinality, not to inclusion-minimality. Write $n(G) = |V(G)|$ and $e(G) = |E(G)|$.

Fischermann, Rautenbach and Volkmann [1] investigated the maximum size of graphs with a unique minimum dominating set. Fraboni and Shank [2] subsequently settled their conjectured size bound when the domination number is two. Koch and Narayan [4] considered the bipartite problem and proposed a bound depending on the order and the domination number. They proved their bound for domination number two and at order $n = 3\gamma$. Erlbacher's public artifacts [3] already give the 13-vertex, 22-edge counterexample, an unbalanced family containing a 14-vertex, 28-edge example, and finite extremal computations. The results below recover those examples, determine the sharp maxima and equality classes at the next two orders, and examine their quantitative edit stability. Koch and Narayan also identify the unique extremal graph at $(n, \gamma) = (10, 3)$ in their Figure 1 discussion.

The order restriction is intrinsic. If D is the unique minimum dominating set of a graph with no isolated vertices, every vertex of D has at least two exterior private neighbours with respect to D . These neighbours are distinct for distinct members of D , and hence $n(G) \geq 3\gamma(G)$. We recall the short argument in Lemma 3.1.

For integers $\gamma \geq 2$ and $r \in \{1, 2\}$, let $\mathcal{B}_{\gamma,r}$ be the class of bipartite graphs with no isolated vertices, of order $3\gamma + r$, whose minimum dominating set is unique and has cardinality γ . Write $\mathcal{B}_\gamma = \mathcal{B}_{\gamma,1}$. No connectedness assumption is imposed in these definitions.

Theorem 1.1. *For every integer $\gamma \geq 2$,*

$$\max_{G \in \mathcal{B}_\gamma} e(G) = \frac{\gamma(\gamma + 7)}{2}.$$

The maximum is attained by a connected graph. For every $\gamma \geq 2$, equality holds if and only if

$$G \cong H(\lceil \gamma/2 \rceil, \lfloor \gamma/2 \rfloor),$$

where $H(p, q)$ is the graph defined in Section 2.

Theorem 1.2. *For every integer $\gamma \geq 2$,*

$$\max_{G \in \mathcal{B}_{\gamma,2}} e(G) = \left\lceil \frac{\gamma^2}{2} \right\rceil + 5\gamma.$$

The equality graphs, up to isomorphism, are exactly

$$\begin{cases} H_2(1, 1), F_0, F_1, & \gamma = 2, \\ H_2(k + 1, k), & \gamma = 2k + 1 \geq 3, \\ H_2(k, k), H_2(k + 1, k - 1), & \gamma = 2k \geq 4. \end{cases}$$

Here H_2 is defined in Section 2, and the two balanced eight-vertex exceptions F_0, F_1 in Section 7. Every equality graph is connected.

For graphs G, H of the same order, define

$$d_{\text{edit}}(G, H) = \min_{\phi: V(H) \rightarrow V(G) \text{ bijective}} |E(G) \triangle \phi(E(H))|.$$

Each edge addition or deletion costs one. The bijection need not preserve the displayed bipartition or a named dominating set.

Theorem 1.3. *Let $\gamma \geq 2$ and $G \in \mathcal{B}_{\gamma,1}$, let $H^* = H(\lceil \gamma/2 \rceil, \lfloor \gamma/2 \rfloor)$, and put $t = \gamma(\gamma + 7)/2 - e(G)$. Then*

$$d_{\text{edit}}(G, H^*) \leq \min\{\gamma(\gamma + 7) - t, 12\gamma t\}.$$

For every $k \geq 2$ and $1 \leq s \leq k - 1$, there is a connected graph $G(k, s) \in \mathcal{B}_{2k,1}$ with exact deficit $t = s$ such that

$$s(k + 1) \leq d_{\text{edit}}(G(k, s), H(k, k)) \leq s(2k + 1).$$

Thus the dependence γt has the correct order for small absolute edge deficit. It cannot generally be replaced by $\gamma\sqrt{t} + t$. More strongly, the connected examples have relative edge deficit tending to zero but normalized edit distance bounded away from zero; the precise limits are given in Section 9.

At order $3\gamma + 1$, Conjecture 1 of [4] specializes to

$$e(G) \leq 2\gamma + 2ab + 2a + 1, \quad a = \lceil \gamma/2 \rceil, \quad b = \lfloor \gamma/2 \rfloor. \quad (1)$$

The summation term in the conjecture is empty at this order. Since

$$\frac{\gamma(\gamma + 7)}{2} - (2\gamma + 2ab + 2a + 1) = b - 1, \quad (2)$$

Theorem 1.1 has the following consequence.

Corollary 1.4. *For every $\gamma \geq 4$, there is a connected bipartite graph on $3\gamma + 1$ vertices with a unique minimum dominating set of cardinality γ that violates (1). The excess is $\lfloor \gamma/2 \rfloor - 1$. In particular, at $\gamma = 4$ the extremal graph has 13 vertices and 22 edges, whereas (1) gives 21.*

The additional vertex explains the change from the threshold case $n = 3\gamma$. It may be adjacent to several members of the unique minimum dominating set. Our extremal graph uses this possibility. The proof of the upper bound keeps track of whether the additional vertex has one or several such neighbours; the equality argument then forces the whole construction.

2 The extremal constructions

Definition 2.1. For integers $p, q \geq 1$ and $r \in \{1, 2\}$, let $H_r(p, q)$ have pairwise distinct vertices

$$X = \{x_i : 1 \leq i \leq p\}, \quad Y = \{y_j : 1 \leq j \leq q\},$$

$$U_i = \{u_i^0, u_i^1\}, \quad V_j = \{v_j^0, v_j^1\}, \quad Z = \{z_1, \dots, z_r\}.$$

Its edges are exactly $x_i u_i^\varepsilon$, $y_j v_j^\varepsilon$, $u_i^\varepsilon v_j^0$, $z_t u_i^\varepsilon$ and $z_t y_j$, for every admissible index and $\varepsilon \in \{0, 1\}$. Write $H(p, q) = H_1(p, q)$, and write z for its single member of Z .

The graph is bipartite, with parts

$$X \cup \bigcup_j V_j \cup Z \quad \text{and} \quad Y \cup \bigcup_i U_i. \quad (3)$$

It is connected: each z_t meets all of $Y \cup \bigcup_i U_i$, each x_i meets U_i , and each member of V_j meets y_j .

Proposition 2.2. *The graph $H_r(p, q)$ has order $3(p + q) + r$, domination number $p + q$, and unique minimum dominating set $X \cup Y$. Its size is*

$$e(H_r(p, q)) = 2(p + q) + 2pq + r(2p + q). \quad (4)$$

Proof. The vertex and edge counts follow from the definition. Clearly $D = X \cup Y$ dominates. Every dominating set K meets each closed neighbourhood $N[x_i] = \{x_i, u_i^0, u_i^1\}$ and each $N[v_j^1] = \{y_j, v_j^1\}$. These $p + q$ sets are pairwise disjoint, so $|K| \geq p + q$.

If $|K| = p + q$, it chooses exactly one member of each of these sets and nothing else. Thus every member of Z , and every v_j^0 , is absent. Choosing u_i^0 instead of x_i leaves its mate undominated: the mate's neighbours are x_i , the members of Z , and the vertices v_j^0 . Hence all x_i are selected and all U_i vertices are absent. Choosing v_j^1 instead of y_j now leaves v_j^0 undominated, forcing all y_j . Thus $K = D$. $\square$

For $a = \lceil \gamma/2 \rceil$ and $b = \lfloor \gamma/2 \rfloor$, direct calculation gives

$$e(H(a, b)) = 2ab + 4a + 3b = \frac{\gamma(\gamma + 7)}{2}, \quad (5)$$

$$e(H_2(a, b)) = 2ab + 6a + 4b = \left\lceil \frac{\gamma^2}{2} \right\rceil + 5\gamma. \quad (6)$$

These graphs supply the connected lower bounds for both theorems.

The subfamily $H_2(p, 1)$ is the two-extra-vertex case of Erlebacher's unbalanced-split construction [3], after interchanging the names of the center sides. In particular, $H_2(3, 1)$ is his 14-vertex example. For $q > 1$, each member of Z meets several centers; this shared-center feature already occurs in his 13-vertex counterexample, which is $H(2, 2)$. We use these constructions to establish sharpness, without claiming priority for those previously reported examples or mechanisms.

3 Private neighbours and local replacements

For a dominating set D and $d \in D$, define

$$\text{epn}(d, D) = \{u \in V(G) \setminus D : N(u) \cap D = \{d\}\}.$$

We use closed neighbourhoods only when square brackets are displayed.

Lemma 3.1. *If G has no isolated vertices and D is its unique minimum dominating set, then $|\text{epn}(d, D)| \geq 2$ for every $d \in D$.*

Proof. If $\text{epn}(d, D) = \{u\}$, the set $(D \setminus \{d\}) \cup \{u\}$ dominates G : u dominates itself and d , while every other vertex outside D retains a neighbour in $D \setminus \{d\}$. This contradicts uniqueness.

If $\text{epn}(d, D) = \emptyset$ and d has a neighbour in D , then $D \setminus \{d\}$ is a smaller dominating set. Otherwise d has no neighbour in D . The no-isolate assumption gives a neighbour u outside D , and $(D \setminus \{d\}) \cup \{u\}$ is again a different dominating set of the same cardinality. All cases are impossible. $\square$

The local lemmas below include every possible edge between the two centers. Their use is governed by the following replacement principle. If an induced subgraph F contains $x, y \in D$ and all other members of D lie outside F , and $D \setminus \{x, y\}$ dominates every vertex outside F , then any dominating set S of F of size at most two extends to $(D \setminus \{x, y\}) \cup S$. Thus, whenever $\{x, y\}$ dominates F , it is its unique dominating set of size at most two. In each application we verify the outside-domination condition explicitly; cross-cell edges cannot obstruct the resulting domination. The six-vertex argument occurs in the proofs of Theorem 1 of [1] and Theorem 12 of [4]; we recall its equality patterns.

Lemma 3.2. *Let F be a bipartite graph with parts $\{x\} \cup V$ and $\{y\} \cup U$, where $|U| = |V| = 2$. Assume that x is adjacent to both vertices of U and y to both vertices of V . If $\{x, y\}$ is the unique dominating set of cardinality at most two, then at most two edges occur among xy and the U – V edges. If exactly two occur, either xy is absent and the U – V edges form a two-edge star, or xy is present and there is exactly one U – V edge.*

Proof. Two disjoint U – V edges give an alternative dominating pair: select opposite endpoints, one from each edge. If xy is absent, three U – V edges contain two disjoint edges. Thus there are at most two, and a two-edge configuration must be a star.

Suppose xy is present. Two U – V edges either are disjoint or form a star. A star centered at $u \in U$ gives the dominating pair $\{x, u\}$; a star centered at $v \in V$ gives $\{y, v\}$. Consequently at most one U – V edge can accompany xy . $\square$

Lemma 3.3. *Let F have the same form as in Lemma 3.2, with $|U| = 2$ and $|V| = s \in \{3, 4\}$. If $\{x, y\}$ is its unique dominating set of size at most two, then at most $2s - 2$ optional edges occur among xy and U – V .*

Proof. If xy is absent and there are at least $2s - 1$ cross-edges, one U row is full and the other has a neighbour v . The full row vertex u and v form an alternative dominating pair. If xy is present, a V vertex adjacent to both members of U gives the alternative $\{y, v\}$. Hence every column has at most one cross-edge, leaving at most $s + 1 \leq 2s - 2$ optional edges in total. $\square$

Lemma 3.4. *In Lemma 3.3 with $s = 3$, equality holds exactly when xy is absent and the cross-edges form a $K_{2,2}$ on U and two members of V , leaving the third column empty.*

Proof. If xy is present at equality, there are three cross-edges and no full column. Each column therefore contains one edge. If one row contains all three, $\{x, u\}$ dominates. Otherwise the row degrees are two and one; the degree-two row vertex together with the other row's neighbour dominates. Both contradict uniqueness.

With xy absent, four cross-edges have row degrees $(3, 1)$ or $(2, 2)$. A full row vertex and a neighbour of the other row rule out $(3, 1)$. For $(2, 2)$ with different row neighbourhoods, select u_0 and the unique member of $N(u_1) \cap V$ outside $N(u_0)$. They dominate the whole cell. Thus both rows meet the same two columns.

Conversely, let c be the empty column. A dominating pair must meet both disjoint sets $N[x] = \{x, u_0, u_1\}$ and $N[c] = \{c, y\}$, and selects nothing else. Choosing a U vertex leaves its mate undominated, forcing x ; then choosing c leaves the active columns undominated, forcing y . No singleton can meet both disjoint sets. $\square$

Lemma 3.5. *Let F have the same form, now with $|U| = |V| = 3$. If $\{x, y\}$ is its unique dominating set of size at most two, then at most six optional edges occur.*

Proof. Suppose at least seven optional edges occur. If xy is absent, the 3 by 3 cross-edge matrix has at most two missing entries. A full row and a full column therefore exist; their vertices form an alternative dominating pair. If xy is present, there are at least six cross-edges. A full row gives $\{x, u\}$, and a full column gives $\{y, v\}$. Otherwise every row and column has degree at most two. Six edges force every degree to equal two, so the missing entries form a perfect matching. The endpoints of any missing edge dominate both centers and every private vertex, again a contradiction. $\square$

These elementary arguments give the optional-edge bounds 2, 4, 6, 6 for private-set sizes $(2, 2), (2, 3), (2, 4), (3, 3)$ respectively. The exact checks in Appendix A are supplementary.

4 One residual vertex

Let $G \in \mathcal{B}_\gamma$, and let D be its unique minimum dominating set. By Lemma 3.1, we may select two exterior private neighbours for each vertex of D . The selected pairs are disjoint and account for 2γ vertices. Since $n(G) = 3\gamma + 1$, exactly one further vertex z remains.

Choose the names of the bipartition classes so that $z \in A$, and write

$$D \cap A = \{x_1, \dots, x_p\}, \quad D \cap B = \{y_1, \dots, y_q\}, \quad p + q = \gamma.$$

Let $U_i \subseteq B$ be the selected private pair of x_i , and $V_j \subseteq A$ the selected private pair of y_j . Set

$$J = N(z) \cap D \subseteq \{y_1, \dots, y_q\}, \quad \ell = |J|.$$

Because D dominates z , we have $\ell \geq 1$, and in particular $q \geq 1$.

Privacy and bipartiteness give a complete list of possible edges:

1. the 2γ selected center-to-private edges;
2. $x_i y_j$ and the edges between U_i and V_j , for each pair (i, j) ;
3. the edges from z to J and to $\bigcup_i U_i$.

In particular, edges inside D have not been discarded: they are precisely the possible edges $x_i y_j$ in the second class.

4.1 The additional vertex has several dominators

Suppose $\ell \geq 2$. For each i, j , consider the induced six-vertex graph

$$G[K_{ij}], \quad K_{ij} = \{x_i, y_j\} \cup U_i \cup V_j.$$

The set $D \setminus \{x_i, y_j\}$ dominates every vertex outside K_{ij} . Each selected private vertex outside the cell retains its own center; the vertex z retains a neighbour in J because only one y -center has been removed.

If $G[K_{ij}]$ had a dominating set S of cardinality at most two other than $\{x_i, y_j\}$, then

$$(D \setminus \{x_i, y_j\}) \cup S$$

would be a dominating set of G of cardinality at most γ , distinct from D when its cardinality is γ . This contradicts minimum cardinality or uniqueness. Thus Lemma 3.2 applies to every cell. The optional edge sets of these cells are disjoint, and $\deg(z) \leq 2p + q$. Consequently

$$e(G) \leq 2\gamma + 2pq + 2p + q. \quad (7)$$

When $p = 0$, the same bound follows directly from the edge list, with an empty sum over cells.

4.2 The additional vertex has one dominator

Suppose $J = \{y_{j_0}\}$. For every $j \neq j_0$, the preceding six-vertex replacement still applies, since y_{j_0} is retained and dominates z . For each i consider instead

$$L_i = \{x_i, y_{j_0}\} \cup U_i \cup V_{j_0} \cup \{z\}.$$

In $G[L_i]$, the center y_{j_0} has the three private vertices $V_{j_0} \cup \{z\}$. Every vertex outside L_i is dominated by $D \setminus \{x_i, y_{j_0}\}$, so the same replacement argument proves that Lemma 3.3 applies.

The optional edges of $G[L_i]$ are $x_i y_{j_0}$ and the edges from U_i to $V_{j_0} \cup \{z\}$. These edge sets are disjoint for different i . The edge $z y_{j_0}$ is counted once separately. Therefore

$$\begin{aligned} e(G) &\leq 2\gamma + 1 + 2p(q - 1) + 4p \\ &= 2\gamma + 1 + 2pq + 2p \\ &\leq 2\gamma + 2pq + 2p + q. \end{aligned} \quad (8)$$

If $p = 0$, only the 2γ fixed private edges and $z y_{j_0}$ occur, so the displayed inequalities remain valid.

4.3 Optimizing the bipartition

Put $d = p - q$. Since $p + q = \gamma$, both cases yield

$$e(G) \leq 2\gamma + 2pq + 2p + q = \frac{\gamma(\gamma + 7)}{2} - \frac{d(d - 1)}{2} \leq \frac{\gamma(\gamma + 7)}{2}. \quad (9)$$

The final inequality holds because d is an integer. Equality in the numerical optimization requires $d \in \{0, 1\}$. The parity of γ then forces

$$p = \lceil \gamma/2 \rceil, \quad q = \lfloor \gamma/2 \rfloor. \quad (10)$$

Together with (5), this proves the extremal-value assertion of Theorem 1.1 for every $\gamma \geq 2$.

5 Equality with one residual vertex

5.1 Domination number at least four

Suppose now that $\gamma \geq 4$ and $e(G) = \gamma(\gamma + 7)/2$. Equation (10) gives $q = \lfloor \gamma/2 \rfloor \geq 2$. The final inequality in (8) loses $q - 1$ edges, so the case $\ell = 1$ cannot give equality. Every inequality in (7) must therefore be an equality. It follows that z is adjacent to every y_j and to every vertex of $\bigcup_i U_i$, and each six-vertex cell has exactly two optional edges.

We first show that D is independent. If $x_i y_j$ were an edge, then

$$(D \setminus \{x_i\}) \cup \{z\}$$

would be a different dominating set of cardinality γ . Indeed, x_i is dominated by the retained y_j , the pair U_i is dominated by z , and every other private pair retains its center. Thus no edge $x_i y_j$ occurs.

Lemma 3.2 now shows that the two edges between U_i and V_j form a star. This star cannot be centered at $u \in U_i$, because

$$(D \setminus \{x_i, y_j\}) \cup \{u, z\}$$

would dominate G . The vertex u dominates x_i and both members of V_j , while z dominates y_j and both members of U_i . All other private pairs retain their centers. Therefore both members of U_i are adjacent to exactly one vertex of V_j ; denote it by $v_j^{(i)}$.

Fix j . Suppose that the selected vertex $v_j^{(i)}$ is not constant as i varies. Choose one vertex $u_i \in U_i$ for every i , and define

$$K = \{z\} \cup \{u_i : 1 \leq i \leq p\} \cup ((D \cap B) \setminus \{y_j\}).$$

Then $|K| = 1 + p + (q - 1) = \gamma$. Every x_i is dominated by u_i ; every vertex in $\bigcup_i U_i$ and every y -center is dominated by z . The two vertices of V_j are dominated because both possible choices of $v_j^{(i)}$ occur. Every other pair V_ℓ retains y_ℓ . Hence K is a second minimum dominating set, a contradiction.

For each j , rename the members of V_j so that their common selected vertex is v_j^0 . The complete edge list is now exactly that of $H(p, q)$. Equation (10) proves the equality assertion in Theorem 1.1 for $\gamma \geq 4$.

5.2 Domination numbers two and three

At equality, (10) gives $q = 1$ and $p = 1$ or 2 . Write y for the sole Y -center and put $W = V_1 \cup \{z\}$. All three vertices of W meet y . The graph has $2p + 3$ fixed center/private edges, and its optional edges partition into the p blocks $x_i y$ and $U_i - W$. Each seven-vertex cell is legitimate for replacement: every outside vertex retains its own center, while z is inside the cell. Equality forces four optional edges in every block.

Lemma 3.4 forbids all center edges and makes each U_i complete to two of the three W columns. For $p = 1$ this is $H(1, 1)$. For $p = 2$, suppose the omitted columns differ. The active column pairs then cover W and have one common member w . Choose $u_i \in U_i$. The set $\{u_1, u_2, w\}$ dominates: the u_i cover x_i and all columns between them, while w covers both U pairs and y . This is a different dominating triple, a contradiction. The omitted columns therefore coincide, and the graph is $H(2, 1)$. This completes Theorem 1.1 for all $\gamma \geq 2$. As noted above, the latter finite equality graph was already identified in [4].

6 Two residual vertices

Let $G \in \mathcal{B}_{\gamma, 2}$ and let D be its unique minimum dominating set. Select two exterior private neighbours of each member of D , leaving two residual vertices z, w . Put

$$D \cap A = X = \{x_1, \dots, x_p\}, \quad D \cap B = Y = \{y_1, \dots, y_q\}, \quad p + q = \gamma,$$

and use $U_i \subseteq B$, $V_j \subseteq A$ for the selected private pairs. We prove the upper bound

$$M_2(\gamma) = \left\lceil \frac{\gamma^2}{2} \right\rceil + 5\gamma.$$

6.1 Residual vertices in the same part

Orient the parts so that $z, w \in A$. Each has a nonempty D -neighbourhood contained in Y , hence $q \geq 1$; $p = 0$ is allowed. Call a residual vertex *singleton-dominated* if it has one neighbour in D . Let $S \in \{0, 1, 2\}$ be the number of such vertices, and let r_j count those whose sole D -neighbour is y_j , so that $\sum_j r_j = S$.

For each i, j , take the cell with centers x_i, y_j , private set U_i , and opposite private set V_j together with the r_j residuals owned by y_j . Every vertex outside this cell remains dominated by $D \setminus \{x_i, y_j\}$: selected private vertices retain their centers; singleton residuals outside have a different retained owner; residuals with at least two D -neighbours retain one, because only one Y -center is removed. Lemmas 3.2 and 3.3 therefore bound its optional edges by $2 + 2r_j$.

These optional edge sets are pairwise disjoint. In addition to the 2γ fixed selected-private incidences, count the S singleton owner edges separately. Each of the remaining $2 - S$ residuals has degree at most $2p + q$: it can meet only Y and the U vertices. There is no edge zw . Consequently

$$\begin{aligned} e(G) &\leq 2\gamma + S + 2pq + 2pS + (2 - S)(2p + q) \\ &= 2\gamma + 2pq + 4p + 2q - S(q - 1) \\ &\leq 2pq + 6p + 4q. \end{aligned} \tag{11}$$

This includes $p = 0$, with no cells. For $d = p - q$, the final expression is

$$\frac{\gamma^2}{2} + 5\gamma + \frac{1}{2} - \frac{(d - 1)^2}{2}. \tag{12}$$

If γ is even, d is even and $(d - 1)^2 \geq 1$; if γ is odd, $(d - 1)^2 \geq 0$. Thus (12) is at most $M_2(\gamma)$.

6.2 Residual vertices in opposite parts

Suppose $z \in A$, $w \in B$, and write

$$J = N(z) \cap D \subseteq Y, \quad K = N(w) \cap D \subseteq X.$$

Both sets are nonempty, so $p, q \geq 1$. The edge zw is now possible and must be counted once.

Both residuals have several dominators. Suppose $|J|, |K| \geq 2$. Each ordinary six-vertex cell is valid: both residuals retain a D -neighbour after one X -center and one Y -center are removed. The cell edges total at most $2pq$. The edges incident with z other than zw number at most $2p + q$, and those incident with w other than zw number at most $2q + p$. Hence

$$e(G) \leq 2\gamma + 2pq + (2p + q) + (2q + p) + 1 = 2pq + 5\gamma + 1. \tag{13}$$

For odd γ , the inequality $2pq \leq \lfloor \gamma^2/2 \rfloor$ gives the result. For even γ , (13) is at most $M_2(\gamma) + 1$. An actual violation would, by integrality, force equality throughout, $p = q$, and saturation of both residual degree bounds. In particular, z meets every Y -center and every U vertex, while w meets every X -center and every V vertex. For any i, j , the set

$$(D \setminus \{x_i, y_j\}) \cup \{z, w\}$$

then dominates. The removed centers are covered by w, z respectively, U_i by z , and V_j by w ; all other private pairs retain their centers. This second dominating γ -set excludes the only possible excess.

Exactly one residual has one dominator. Interchange sides if necessary so that $J = \{y_{j_0}\}$ and $|K| \geq 2$. For $j \neq j_0$, use ordinary six-vertex cells. For $j = j_0$, include z and use a $(2, 3)$ private cell. Outside every such cell, w retains an X -neighbour; any residual not included keeps its owner. Thus the optional blocks contribute at most $2p(q-1) + 4p$. Count zy_{j_0} once, the other incidences of w by $2q + p$, and zw at most once. This gives

$$e(G) \leq 2\gamma + 1 + 2p(q-1) + 4p + (2q + p) + 1 = 2pq + 5\gamma + 2 - q. \quad (14)$$

If $q \geq 2$, this is at most $M_2(\gamma)$. If $q = 1$, then $p \geq 2$ because $|K| \geq 2$, so $\gamma \geq 3$. The difference between $M_2(\gamma)$ and the right side is $\lceil \gamma^2/2 \rceil - 2\gamma + 1$, which is zero at $\gamma = 3$ and positive for $\gamma \geq 4$.

Both residuals have one dominator. Write $J = \{y_{j_0}\}$ and $K = \{x_{i_0}\}$. Partition optional edges using $(2, 2)$ cells for $i \neq i_0, j \neq j_0$; $(3, 2)$ cells including w for $i = i_0, j \neq j_0$; $(2, 3)$ cells including z for $i \neq i_0, j = j_0$; and one $(3, 3)$ cell containing both residuals for $i = i_0, j = j_0$. Every residual whose owner is removed is included in the cell, and every residual outside retains its owner. All selected private vertices outside also retain their centers. Thus all local uniqueness hypotheses hold.

The edge zw belongs only to the $(3, 3)$ cell. The two residual owner edges are counted once each as fixed incidences, and the optional blocks are disjoint. The local bounds give

$$\begin{aligned} e(G) &\leq 2\gamma + 2 + 2(p-1)(q-1) + 4(q-1) + 4(p-1) + 6 \\ &= 2pq + 4\gamma + 2 \\ &\leq \lceil \gamma^2/2 \rceil + 4\gamma + 2 \leq M_2(\gamma), \end{aligned} \quad (15)$$

using $\gamma \geq 2$. This covers the small case $p = q = 1$ as well. Together with (6), all cases prove the extremal value in Theorem 1.2.

6.3 Nonisomorphic equality graphs for even domination number

Let $\gamma = 2k \geq 4$. Both $H_2(k, k)$ and $H_2(k+1, k-1)$ attain the maximum by (4). Their bipartition size pairs are respectively $\{3k, 3k+2\}$ and $\{3k+1, 3k+1\}$. A connected bipartite graph has its bipartition determined up to interchange, so the graphs are nonisomorphic. The complete equality classification is proved next.

7 Equality with two residual vertices

Assume $e(G) = M_2(\gamma)$. We retain the private-pair notation of Section 6. No connectedness assumption is used in the classification.

7.1 Opposite-side residuals above domination number two

Suppose first that both residuals have several dominators. For odd γ , equality in (13) forces a balanced split and saturation of all residual incidences. Replacing any x_i, y_j by z, w gives another dominating γ -set, as in the upper-bound proof.

For even γ , equality requires $p = q = k \geq 2$; any other split reduces $2pq$ by at least two and makes (13) strictly smaller than $M_2(\gamma)$. There is exactly one total deficit in that bound. All cell deficits and residual-incidence deficits are nonnegative integers. Thus at most one of the allowable residual edges is missing. If it is incident with an X -center or one of its private vertices, choose a different X -center to remove; if it is incident with a Y -center or its private pair, choose a different Y -center. Such choices exist because $k \geq 2$. If the deficit is a cell edge or zw , choose any pair. The set

$$(D \setminus \{x_i, y_j\}) \cup \{z, w\}$$

then dominates: the removed centers and their pairs have all needed residual incidences, and other private pairs retain their centers. A missing edge to a retained center or its pair is harmless. This again contradicts uniqueness.

Suppose exactly one residual is singleton-dominated, with $J = \{y_{j_0}\}$ and $|K| \geq 2$. The difference between the target and the right side of (14) is

$$\lceil \gamma^2/2 \rceil - 2pq + q - 2.$$

For $q \geq 2$ this vanishes only at $p = q = 2$: odd γ contributes at least one in the first two terms, and even γ requires $p = q$ and $q = 2$. For $q = 1$, it vanishes only at $\gamma = 3, p = 2$. In either possible split, the two special seven-cells on x_1, y_{j_0} and x_2, y_{j_0} are saturated. By Lemma 3.4, each U_i is complete to two members of $W = V_{j_0} \cup \{z\}$. Choose a common member c of these two two-element subsets of a three-set. Saturation also makes w adjacent to all X -centers, all V vertices, and z . Hence

$$\{w, c\} \cup (Y \setminus \{y_{j_0}\})$$

is a dominating set of size $q + 1 = \gamma - 1$. The vertex c covers both U pairs and y_{j_0} ; w covers X , all V and z ; the other Y -centers are selected. This contradicts the domination number.

Finally, if both residuals are singleton-dominated, the gap in (15) is at least $\gamma - 2 > 0$ for $\gamma \geq 3$. Thus every equality graph with $\gamma \geq 3$ has same-side residuals. These arguments apply to any selected exterior-private pairs, not a presumed favourable decomposition.

7.2 Same-side equality with at least two opposite centers

In (11), equality forces $S = 0$, complete residual incidences to every Y -center and every U vertex, and two optional edges in every six-cell. A center edge $x_i y_j$ would allow replacing x_i by z , so all center edges are absent. Each block is a two-edge star. A star centered at $u \in U_i$ would allow replacing x_i, y_j by u, z ; the other residual retains a Y -neighbour since $q \geq 2$. Thus every block is a column star.

For a fixed j , its chosen column cannot vary with i . Otherwise

$$\{z\} \cup \{u_i : u_i \in U_i, 1 \leq i \leq p\} \cup (Y \setminus \{y_j\})$$

would be another dominating γ -set, by the argument in Section 5. In this setting w is also covered, since it retains a Y -neighbour. Relabelling the common column of each V_j as v_j^0 gives precisely $H_2(p, q)$.

Equality in (12) requires $p - q = 1$ for odd γ and $p - q \in \{0, 2\}$ for even γ . These are the splits stated in Theorem 1.2.

7.3 Same-side equality with one opposite center

The allowed splits now give $p = 1, 2, 3$, corresponding to $\gamma = 2, 3, 4$. Both residuals have the sole center y as owner. Put $W = V_1 \cup \{z, w\}$, of size four. Every $(2, 4)$ cell must have six optional edges.

Its equality pattern has no center edge and consists of $K_{2,3}$ on U and three columns of W . Indeed, if xy is present, uniqueness excludes a full column, leaving at most four cross-edges and five optional edges. With xy absent, six cross-edges have row degrees $(4, 2)$ or $(3, 3)$. A full row and a neighbour of the other row exclude $(4, 2)$. Different three-element row neighbourhoods give an alternative pair by selecting one row vertex and the other's neighbour outside that row. Thus the two rows have the same three neighbours. Conversely, the unused column and x have disjoint closed neighbourhoods forcing the center pair, just as in Lemma 3.4.

Let c_i be the omitted column for U_i . If the omissions differ, the active columns cover W , and some column c is active in every row because $p \leq 3 < 4$. Choosing one $u_i \in U_i$, the set $\{c, u_1, \dots, u_p\}$ is another dominating $(p + 1)$ -set: it covers all centers, all U vertices, and all of W . Hence the omitted column is common, and relabelling the three active columns gives $H_2(p, 1)$.

7.4 The three equality graphs at domination number two

The remaining possibility has opposite-side residuals and $p = q = 1$. The entire graph is the $(3, 3)$ private cell. Write $U = \{u_0, u_1, u_2\}$ and $V = \{v_0, v_1, v_2\}$, with fixed edges xU, yV . There must be six optional edges.

If xy is absent, there are six cross-edges. With no full row or column, every row and column would have degree two, and a missing-edge pair would dominate. By symmetry assume a full row. There is no full column, or its vertex and the full-row vertex would dominate. The other three cross-edges occupy the three columns once each. Row degrees $(2, 1)$ give a dominating pair, so they must be $(3, 0)$. This yields the graph F_0 : no edge xy , and the six cross-edges $\{u_0, u_1\} \times V$.

If xy is present, the five cross-edges have no full row or column; the row and column degree lists are $(2, 2, 1)$. Their graph has maximum degree two, no isolates, six vertices and five edges. Its only component types are P_6 and $C_4 \cup K_2$. In the path $u_0v_0u_1v_1u_2v_2$, the pair $\{u_2, v_0\}$ dominates the cell. Thus only the second type remains. This gives F_1 : the edge xy , the four cross-edges $\{u_0, u_1\} \times \{v_0, v_1\}$, and u_2v_2 .

Both graphs have unique minimum dominating pair $\{x, y\}$. In F_0 , a dominating pair meets the disjoint sets $N[u_2] = \{u_2, x\}$ and $N[y] = \{y, v_0, v_1, v_2\}$. Selecting u_2 leaves an active U vertex or a V vertex undominated, forcing x ; a V vertex instead of y then leaves the other V vertices undominated. In F_1 , a pair with one center other than $\{x, y\}$ cannot cover the opposite private set. A pair inside U misses y , and one inside V misses x . An alternative $\{u, v\}$ with $u \in U, v \in V$ would require both cross-degrees to be two and uv to be a nonedge. All such degree-two vertices lie in the C_4 component, where every opposite pair is adjacent. A singleton cannot dominate a graph with these part sizes.

Together with the same-side graph $H_2(1, 1)$, these are exactly the three classes in Table 1. The degree sequences separate F_0, F_1 , and their balanced bipartitions distinguish them from $H_2(1, 1)$. For larger even γ , the two listed H_2 graphs are nonisomorphic by Section 6. All listed graphs are connected and attain the bound, completing Theorem 1.2.

Graph	Bipartition sizes	Sorted degree sequence
$H_2(1, 1)$	3, 5	1, 2, 3, 3, 3, 4, 4, 4
F_0	4, 4	1, 3, 3, 3, 3, 3, 4, 4
F_1	4, 4	2, 2, 3, 3, 3, 3, 4, 4

Table 1: The three eight-vertex equality graphs, each with twelve edges.

8 Stability in terms of absolute edge deficit

Let $G \in \mathcal{B}_{\gamma, 1}$ and put $t = \gamma(\gamma + 7)/2 - e(G)$. Theorem 1.1 shows $t \geq 0$; if $t = 0$, its equality statement gives $d_{\text{edit}}(G, H^*) = 0$. Suppose henceforth $t \geq 1$. Selecting two exterior private neighbours per center leaves $z \in A$, with p, q, U_i, V_j as before.

8.1 An exact deficit decomposition

First suppose z has at least two neighbours in D . Let $k_{ij} \leq 2$ be the optional-edge count of the six-cell on x_i, y_j . Define

$$B = \frac{(p - q)(p - q - 1)}{2}, \quad L = \sum_{i, j} (2 - k_{ij}), \quad R = 2p + q - \deg(z).$$

Each is a nonnegative integer. Complete edge accounting gives the exact identity

$$t = B + L + R. \tag{16}$$

Here B measures center-split imbalance, L the cell deficits, and R the missing allowable residual incidences.

For an intermediate template, allow $p = 0$ and write $T(p, q)$ for the same adjacency rule as $H(p, q)$ with an empty X collection when necessary. We still have $q \geq 1$. The final balanced template has both center sides nonempty.

Call row i bad if z misses a member of U_i , and column j bad if z misses y_j . Let their counts be b_X, b_Y , and let R_U, R_Y count the corresponding missing incidences. Then

$$b_X \leq R_U, \quad b_Y \leq R_Y, \quad R_U + R_Y = R.$$

On a good row, no center edge $x_i y_j$ can occur: replacing x_i by z would cover its private pair and leave x_i dominated by the retained y_j .

A saturated block in a good row and good column is therefore a cross-edge star. It cannot be centered at $u \in U_i$, since replacing x_i, y_j by u, z would dominate. Thus it is a column star. The selected column is common to all good saturated rows at a fixed good j . Otherwise choose two good rows i, h with opposite selections. For $u_i \in U_i, u_h \in U_h$, the set

$$(D \setminus \{x_i, x_h, y_j\}) \cup \{z, u_i, u_h\}$$

would be another dominating γ -set. The two selected private vertices cover their centers and both members of V_j , while z covers their private pairs and y_j ; all other private pairs retain their centers. This proves coherence even in the presence of bad rows elsewhere.

Choose that common column for each good j , or an arbitrary column when there is no good saturated row; choose arbitrary columns for bad j . These choices define a labelled $T(p, q)$. Its fixed private edges already agree with those of G . Add the R missing residual incidences. Every block touching a bad row or column costs at most four edits, since each graph has at most two optional edges there. The same bound applies to each deficient good block, of which there are at most L . All remaining blocks agree. Thus

$$\begin{aligned} d_{\text{edit}}(G, T(p, q)) &\leq R + 4L + 4(qb_X + pb_Y) \\ &\leq (4\gamma + 1)R + 4L \\ &\leq (4\gamma + 1)t. \end{aligned} \tag{17}$$

The displayed repair uses a common labelling, so allowing arbitrary bijections only reduces its cost. The argument includes $p = 0$.

8.2 Balancing the template

Put $a = \lceil \gamma/2 \rceil$, $b = \lfloor \gamma/2 \rfloor$ and $h = |p - a|$. Writing $p = a + m$ gives $B = 2m^2 - m$ for even γ and $B = 2m^2 + m$ for odd γ . For integer m , both are at least m^2 . Hence

$$h^2 \leq B \leq t. \tag{18}$$

Move h center/private triples to the other role, keeping the residual and all unchanged triples fixed. Keep selected-column labels on unchanged Y -triples. This is a bijection to $T(a, b) = H^*$, and the two templates agree on all edges whose endpoints both lie outside the moved triples.

If $p > a$, a moved X -triple has degree sum $2q + 6$ in the old template and $2a + 5$ in its new Y -role. If $p < a$, a moved Y -triple has degree sum $2p + 5$, and its new X -role has sum $2b + 6$. In either case the sum of old and new degrees is $2\gamma - 2h + 11$. Counting each changed edge by an incident moved vertex, with repeated counting allowed for an upper bound, gives

$$d_{\text{edit}}(T(p, q), H^*) \leq h(2\gamma - 2h + 11) \leq (2\gamma + 11)\sqrt{t}. \tag{19}$$

By the triangle inequality, (17) and (19) yield

$$d_{\text{edit}}(G, H^*) \leq (4\gamma + 1)t + (2\gamma + 11)\sqrt{t} \leq (6\gamma + 12)t \leq 12\gamma t,$$

using the integer condition $t \geq 1$ and $\gamma \geq 2$.

8.3 A residual vertex with one owner

If z has one D -neighbour, (8) gives

$$t \geq B + q - 1 = \frac{(p - q - 1)^2 + \gamma - 3}{2} \geq \left\lfloor \frac{\gamma - 2}{2} \right\rfloor,$$

where the last step uses parity. For $\gamma = 2, 3$ and $t \geq 1$, the trivial distance bound $d_{\text{edit}}(G, H^*) \leq \gamma(\gamma + 7)$ is at most $12\gamma t$. For $\gamma \geq 4$, use

$$\gamma + 7 \leq 12 \left\lfloor \frac{\gamma - 2}{2} \right\rfloor$$

to obtain the same conclusion. This proves the linear-deficit bound for every graph in the class, without connectedness. The additional bound

$$d_{\text{edit}}(G, H^*) \leq e(G) + e(H^*) = \gamma(\gamma + 7) - t$$

is immediate. Together they establish the upper assertion of Theorem 1.3.

9 Connected obstructions to stronger edit stability

Let $k \geq 2$ and $1 \leq s \leq k - 1$. Starting from $H(k, k)$, modify each of the rows $i = 1, \dots, s$ by deleting the k edges $u_i^0 v_j^0$, adding the k edges $x_i y_j$, and deleting $z u_i^0$. Call the result $G(k, s)$. The same private vertex loses both its selected-column edges and its residual incidence, becoming a leaf at x_i . Its mate u_i^1 retains all its edges.

Every change respects the original bipartition. Connectivity is preserved: z meets all y_j , every unmodified U vertex and every u_i^1 ; each x_i meets u_i^1 , each V vertex meets its owner, and the modified leaves meet x_i . There are no isolated vertices. The order is $6k + 1$ and the edge count is $2k^2 + 7k - s$.

9.1 The unique minimum dominating set

Every dominating set meets the following pairwise disjoint closed neighbourhoods:

$$\{x_i, u_i^0\} \quad (i \leq s), \quad \{x_i, u_i^0, u_i^1\} \quad (i > s), \quad \{y_j, v_j^1\} \quad (1 \leq j \leq k).$$

They are respectively the closed neighbourhoods of the modified leaves, the unmodified X -centers, and the V leaves. There are $2k$ of them, so every dominating set has at least $2k$ vertices. The set $D = X \cup Y$ dominates and has that size.

A dominating $2k$ -set selects exactly one member of each forced set and nothing else. Thus z , all v_j^0 , and all modified-row u_i^1 are absent. In an unmodified row, selecting a U vertex leaves its mate undominated. In a modified row, selecting u_i^0 instead of x_i leaves u_i^1 undominated, since its neighbours are x_i, z and the v_j^0 . Therefore every x_i is selected. Selecting v_j^1 instead of y_j then leaves v_j^0 undominated. Hence D is the unique minimum dominating set, and $G(k, s) \in \mathcal{B}_{2k, 1}$ has exact deficit s .

9.2 A degree lower bound for every vertex bijection

For any bijection between two graphs,

$$\sum_v |\deg_G(v) - \deg_H(\phi^{-1}(v))| \leq 2|E(G) \triangle \phi(E(H))|.$$

The minimum absolute-difference sum between two degree multisets is achieved by pairing their sorted sequences. Indeed, for $a \leq a'$ and $b \leq b'$, uncrossing follows from

$$|a - b| + |a' - b'| \leq |a - b'| + |a' - b|.$$

Thus half the sorted degree L^1 distance is a lower bound on the edit distance under arbitrary relabelling.

The degree groups of $H(k, k)$, as (degree, multiplicity), are

$$(1, k), (2, k), (3, k), (k + 2, 2k), (2k + 1, k), (3k, 1).$$

For $G(k, s)$ they are

$$(1, k + s), (2, k - s), (3 + s, k), (k + 2, 2k), (2k + 1 - s, k), (3k - s, 1).$$

These groups are nondecreasing for $1 \leq s \leq k - 1$; equalities between groups do not affect the sorted-position calculation. The L^1 difference is $s + ks + ks + s = 2s(k + 1)$. Therefore

$$s(k + 1) \leq d_{\text{edit}}(G(k, s), H(k, k)) \leq s(2k + 1). \quad (20)$$

The upper bound is the explicit modification under the identity labelling: $s(k + 1)$ deletions and sk additions. This completes Theorem 1.3.

9.3 Failure of square-root and dense edit stability

Take $k = s^2$ with $s \geq 2$. The lower bound in (20) is $s(s^2 + 1)$, whereas $\gamma\sqrt{t} + t = 2s^2\sqrt{s} + s$. Their ratio tends to infinity. Hence no absolute constant can bound every graph in the class by $O(\gamma\sqrt{t} + t)$ edits.

For a stronger asymptotic consequence, take $s = \lfloor k/2 \rfloor$. Then $s/(2k^2 + 7k) \rightarrow 0$, so the relative edge deficit tends to zero, while

$$\liminf_{k \rightarrow \infty} \frac{d_{\text{edit}}(G(k, \lfloor k/2 \rfloor), H(k, k))}{(6k + 1)^2} \geq \frac{1}{72}.$$

Thus near-optimal edge density does not imply dense edit-distance closeness, even for connected graphs. In contrast, Theorem 1.3 guarantees normalized edit closeness when $t = o(\gamma)$. A single missing residual incidence in the construction permits changes across an entire row of cells, explaining the linear rather than square-root dependence on the absolute deficit.

10 Source comparison and further questions

At order $3\gamma + 1$, Theorem 1.1 exceeds the conjectured bound by $b - 1$, as in (2). At order $3\gamma + 2$, the same conjecture specializes to $2\gamma + 2ab + 4a + 2$. Its summation tail is again zero, both with the printed cutoff and its construction-consistent correction. Theorem 1.2 exceeds this expression by

$$(2\gamma + 2ab + 4a + 2b) - (2\gamma + 2ab + 4a + 2) = 2(b - 1).$$

Thus the two sharp boundary families violate the proposed bound for every $\gamma \geq 4$. The individual 13-vertex and 14-vertex counterexamples, and the cutoff issue, were already reported in [3]. Neither theorem claims a smallest counterexample over all orders.

For comparison, the published unbalanced-family expression in [3] at two extra vertices is $2\gamma + 2p'q' + 2(2q' + 1)$, with $p' + q' = \gamma$. When $p' = q, q' = p$, its count differs from (4) by $2(q - 1)$. At $\gamma = 5$, its maximum over positive splits is 36, whereas Theorem 1.2 gives 38, attained by $H_2(3, 2)$. This compares the explicit formulas, without asserting priority over every other possible construction.

Both boundary equality classes are now completely determined. Theorem 1.3 shows that exact rigidity does not imply dense edit stability. It remains useful to distinguish other metrics, such as removal of a small vertex set, or additional assumptions, such as independence of the unique minimum dominating set. No theorem for those variants is asserted here. Nor does the present local argument supply a general bound at orders $3\gamma + r$ with $r \geq 3$: the residual vertices’ simultaneous domination obligations must still be retained.

A Supplementary exact checks

Independent exact checkers enumerate all 1696 optional-edge patterns on xy and $U \times V$, testing every vertex subset of size at most two. Table 2 records their results. These computations supplement the analytic proofs.

(a, b)	Optional patterns	Admissible patterns	Maximum optional edges
$(2, 2)$	32	14	2
$(2, 3)$	128	58	4
$(2, 4)$	512	254	6
$(3, 3)$	1024	548	6

Table 2: Admissibility means that $\{x, y\}$ is the only dominating set of size at most two.

The first-boundary low- γ equality reduction was checked on all 128 seven-vertex and 16 384 ten-vertex skeletons, including subextremal patterns. Each has three labelled equality patterns. For the second boundary, all 15 maximum $(3, 3)$ patterns form the two balanced eight-vertex classes, of labelled multiplicities 6 and 9, under independent full-graph isomorphism checks. Among the 164 reduced same-side patterns through $\gamma = 5$, exactly 20 survive, four at each tested split; those survivors map to the stated H_2 representatives. The others have alternative dominating sets. All 9 and 576 mixed-owner candidates at $\gamma = 3, 4$ have the smaller dominating set used in the proof. These are reduced-skeleton checks, not unrestricted graph censuses or counts of isomorphism classes.

All subsets were tested for the ten $H_2(p, q)$ graphs with $p, q \geq 1$ and $2 \leq p + q \leq 5$; H_1 was checked through $\gamma = 6$. Explicit isomorphisms recover the prior 13- and 14-vertex certificates in [3]. Stability checks independently verify domination, forced neighbourhoods, sorted degrees and 819 balancing bijections through $\gamma = 40$; reduced-graph checks also confirm the repair step. The accompanying package supplies the checkers, results, source hashes and independent audits. No Lean formalization is claimed.

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